

# SyriaTel Customer Churn Prediction

*A Machine Learning Approach to Reducing Customer Loss*

3,333

Customers Analyzed

14.5%

Churn Rate

4

Models Built

# Business & Data Understanding



## Stakeholder

SyriaTel Customer Retention Team needs to identify at-risk customers before they leave.



## The Problem

14.5% of customers churn — losing recurring revenue and requiring costly re-acquisition.



## Our Solution

Train a machine learning classifier to predict churn probability for every customer.



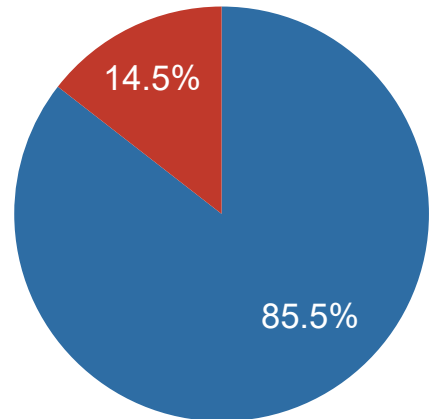
Why Machine Learning? ML captures complex multi-variable patterns (e.g. high charges + international plan + service calls) that simple rules cannot.

# About the Dataset

## Dataset at a Glance

| Category           | Key Features                        |
|--------------------|-------------------------------------|
| Account Info       | state, account length, area code    |
| Plan Info          | international plan, voice mail plan |
| Day Usage          | total day minutes, calls, charge    |
| Evening Usage      | total eve minutes, calls, charge    |
| Night Usage        | total night minutes, calls, charge  |
| Intl Usage         | total intl minutes, calls, charge   |
| Support            | customer service calls              |
| <b>TARGET</b>      | <b>churn (True / False)</b>         |
| 3,333<br>Customers | 20<br>Features                      |
|                    | 0<br>Missing Values                 |

## Target Class Balance



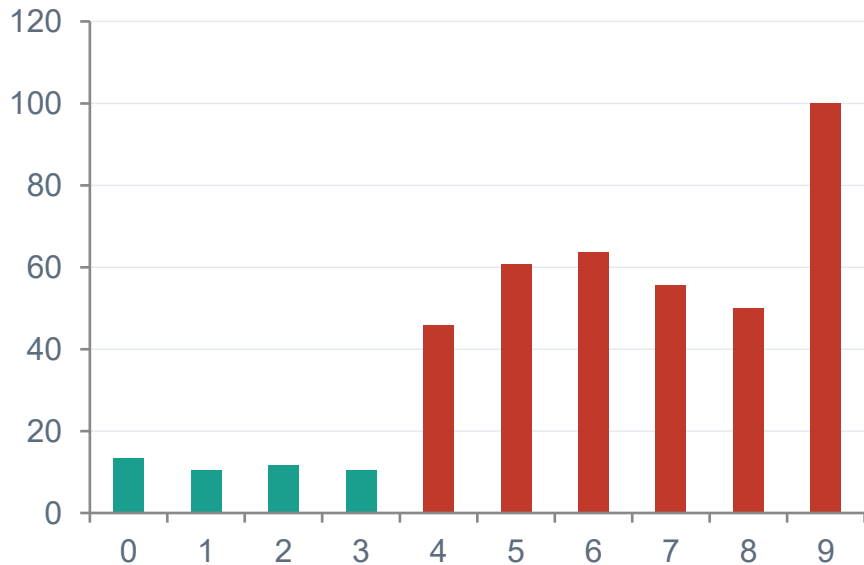
■ No Churn (85.5%) ■ Churn (14.5%)



Class Imbalance — only 14.5% churn.  
Addressed with `class_weight='balanced'`.

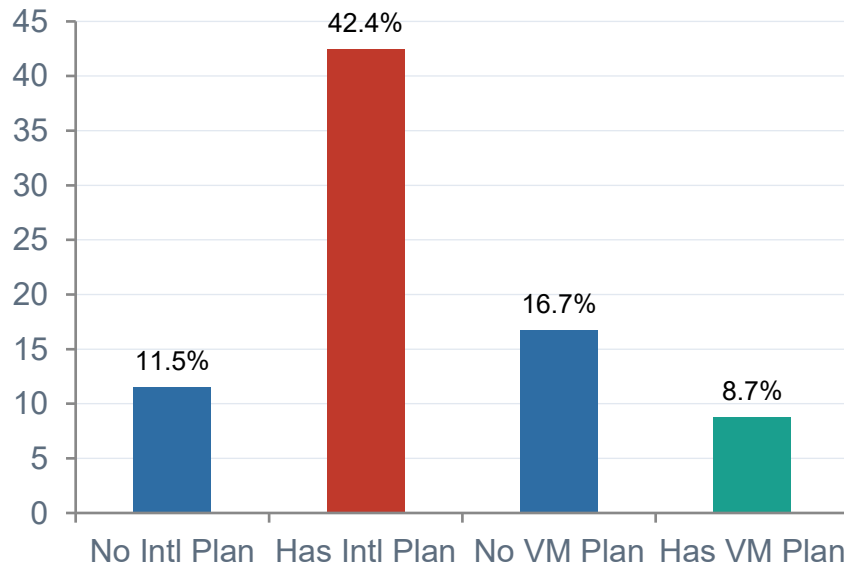
# Key Findings from the Data (EDA)

## Churn Rate by Customer Service Calls



Churn spikes sharply at 4+ service calls (45.8% → 60.6% → 63.6%). Up to 100% at 9 calls.

## Churn Rate by Plan Type



International plan subscribers churn at 42.4% vs 11.5% for others — nearly 4x higher.



Voice mail plan holders churn less (8.7%) — indicating more loyal/engaged customers.

# Our Modeling Approach

1

## Data Prep

Encode categories  
Scale features  
80/20 stratified split  
No data leakage

2

## Logistic Regression

Baseline model  
Simple & interpretable  
Sets our benchmark

3

## Decision Tree

Default + Tuned via  
GridSearchCV  
Non-linear patterns

4

## Random Forest

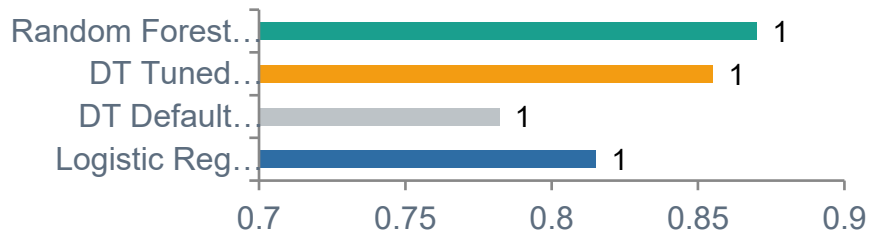
200 trees ensemble  
Best overall performer  
Final model ✓

Primary Metric: RECALL — missing a churner (False Negative) costs far more than a mistaken retention offer to a loyal customer.

# Model Performance Results

| Model                          | Recall | F1 Score | ROC-AUC | Notes                                               |
|--------------------------------|--------|----------|---------|-----------------------------------------------------|
| Logistic Regression (Baseline) | 0.732  | 0.472    | 0.815   | Good recall; lower precision due to class imbalance |
| Decision Tree (Default)        | 0.629  | 0.626    | 0.782   | Overfits: Train Recall=1.000, depth=18              |
| Decision Tree (Tuned)          | 0.701  | 0.571    | 0.855   | Better AUC; still some overfitting (depth=15)       |
| ✓ Random Forest (Final)        | 0.742  | 0.689    | 0.870   | Best overall: highest Recall, F1 & AUC              |

## ROC-AUC Score Comparison

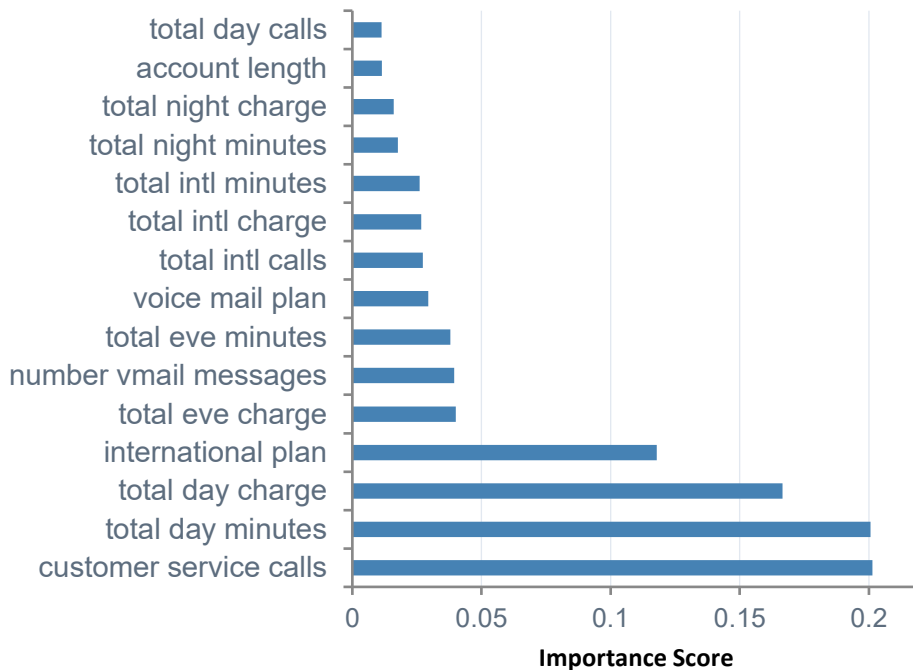


## Why Random Forest?

- ✓ Highest Recall (0.742) — catches 74.2% of churners
- ✓ Best F1 Score (0.689) — precision & recall balance
- ✓ Highest ROC-AUC (0.870) — best discrimination overall
- ✓ Ensemble of 200 trees reduces overfitting

# What Drives Churn? — Feature Importance

## Top 15 Feature Importances — Random Forest



## Business Interpretation



### #1 & #2 Calls + Day Minutes (40.2%)

Two related signals — heavy users making many calls are most at-risk. Check if billing frustration is the root cause.



### #3 Total Day Charge (16.7%)

High bill customers are most likely to shop around. They need a proactive bundle offer before they start comparing.



### #4 International Plan (11.8%)

42.4% churn vs 11.5% average. These subscribers are nearly 4x more likely to leave — urgent review needed.

# Final Model — Confusion Matrix Explained

|                  | Predicted: No Churn                                        | Predicted: Churn                                             |
|------------------|------------------------------------------------------------|--------------------------------------------------------------|
| Actual: No Churn | <b>530</b><br>True Negatives<br>(Correctly kept customers) | <b>40</b><br>False Positives<br>(Loyal flagged as churn)     |
| Actual: Churn    | <b>25</b><br>False Negatives<br>(Churners we MISSED ⚠)     | <b>72</b><br>True Positives<br>(Churners correctly caught ✓) |



## What This Means:

**72 / 97**

Churners correctly caught (74.2% Recall)

**25**

Churners we missed — the key cost to minimize

**40**

False alarms — low cost (retention offer sent to loyal customers)

**90.0%**

Overall accuracy across all 667 test customers



# Recommendations

01

## Flag Customers After 3+ Service Calls

Churn spikes to 45.8% at 4 calls and exceeds 60% at 5+. Auto-escalate on the 3rd call with a senior agent or retention offer.

02

## Urgently Review International Plan Pricing

42.4% churn vs 11.5% baseline — a 4× difference. Proactively contact all international plan subscribers with a pricing audit.

03

## Offer Bundles to High Day-Usage Customers

Total day charge is the 3rd most important driver (16.7%). Offer personalized upgrades before heavy users start comparing competitors.

04

## Deploy Monthly Churn Scoring

Run the Random Forest model monthly on all active customers. Flag those with >60% predicted churn probability for retention outreach.

# Limitations & Next Steps

## ⚠ Model Limitations

- Black-box nature: hard to explain individual predictions
- Class imbalance (14.5%) may cause model to miss rare patterns
- Trained on historical data — must be retrained regularly
- Multicollinearity: day minutes & charge are perfectly correlated ( $r \approx 1.0$ )

## → Next Steps

- Try XGBoost / LightGBM for potentially higher performance
- Apply SMOTE oversampling to better handle class imbalance
- Work with retention team to calibrate decision threshold
- Refresh model quarterly as customer behavior evolves

# Thank You

## *Questions & Discussion*

**0.870**

ROC-AUC

**0.689**

F1 Score

**0.742**

Recall

**74.2%**

Churners Caught

*Final Model: Tuned Random Forest (n=200, max\_depth=5) | Primary Metric: Recall | Author: Joel Muoki Andrew*